Inquiry
An Interdisciplinary Journal of Philosophy

ISSN: 0020-174X (Print) 1502-3923 (Online) Journal homepage: https://www.tandfonline.com/loi/sinq20

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**To cite this article:** Carl Posy (2019): The infinite, the indefinite and the critical turn: Kant via Kripke models, Inquiry, DOI: 10.1080/0020174X.2019.1651095

**To link to this article:** https://doi.org/10.1080/0020174X.2019.1651095

Published online: 15 Oct 2019.

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INQUIRY
https://doi.org/10.1080/0020174X.2019.1651095

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# The infinite, the indefinite and the critical turn: Kant via Kripke models

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**ABSTRACT**
This paper aims to show that intuitionistic Kripke models are a powerful tool for interpreting Kant's 'Critical Philosophy'. Part I reviews some old work of mine that applies these models to provide a reading of Kant's second antinomy about the divisibility of matter and to answer several attacks on Kant's antinomies. But it also points out three shortcomings of that original application. First, the reading fails to account for Kant's second antinomy claim that matter is divisible 'ad infinitum' and his first antinomy claim that empirical space extends only 'ad indefinitum'. Secondly, in conformity with the 'assertability' heuristics for Kripke models, it attributes an assertability semantics to Kant. In fact, it attributes a pair of assertability semantics to Kant, but neither of these accords with modern assertabilism. And finally, one must wonder the propriety of using contemporary assertability and contemporary formal logic to save an eighteenth-century text. Part II goes historical to solve the problems about assertability. It outlines the Descartes–Leibniz dialectic about the infinite/indefinite distinction and demonstrates how each position reflects larger philosophical positions. It then demonstrates that Kant's two versions of assertability and his version of the infinite/indefinite dichotomy emerge naturally from his attempts (pre-critical and then critical) to resolve a tension in Leibniz's philosophy. In light of this demonstration, Part III adapts the Kripke model interpretation so that it does reflect that dichotomy. It then refines the Kripke semantics in order to model some of Kant's signature critical doctrines, and it shows how Kant's critical version of the infinite/indefinite distinction differs from all those of his predecessors and from his own pre-critical version. It also addresses the question of using contemporary logical machinery to interpret Kant. Part IV turns the tables and uses what we have learned about Kant and modern semantics in order to address a pair of issues in the contemporary critical foundations of logic.

**ARTICLE HISTORY** Received 7 June 2019; Accepted 7 June 2019

**KEYWORDS** Kant; Leibniz; Descartes; Kant's antinomies; infinite/indefinite distinction; Kripke models; intuitionistic logic; assertability

I thank the editors for inviting me to contribute to this issue on critical views of logic. Kant invented the *critical* philosophy. He fashioned its

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doctrines (Understanding versus Reason, synthetic *a priority*, transcendental idealism, empirical realism) into a philosophy designed to replace the dogmatic realism of his predecessors and of his own 'pre-critical' thought. His antinomies used *logic* to reduce that realism *ad absurdum*. The realist of the first antinomy says that the physical world is either finite or infinite in extent and yet must deny each disjunct; and similarly the second antinomy realist affirms that there are indivisible finite physical atoms or that matter is infinitely divisible (without atoms), and again denies both. Transcendental idealism (a purely critical doctrine) escapes the *reductio* because, for it, the world is neither finite nor infinite in extent, and there is no infinite division of matter, nor are there material atoms. In both antinomies, the idealist has only continuing series: a series of ever more distant regions, a telescoping series of ever-finer divisions. This, we may say, is the first 'critical' use of logic.

Part I below summarizes old work of mine using intuitionistic Kripke models to explicate that critical use of logic and to defuse some venerable attacks on Kant's text. I highlight a striking technical analogy between an intuitionistic Kripke model and the structure of Kant's idealist picture. I also connect Kant's arguments to Kripke's 'assertabilist' heuristics for his intuitionistic model theory.[1]

But, of course, one must question the propriety of applying assertabilism to save an eighteenth-century text. Moreover, Kant's assertabilism comes in two forms, both of which seem ontologically off-kilter, and one of which actually goes with classical, not intuitionistic, logic.

Worse yet, my old interpretation does nothing to solve a historical anomaly about Kant's idealist version of infinity: Kant says that the second antinomy series of divisions goes on '*ad infinitum*', while the first antinomy's series of further regions continues only '*ad indefinitum*'. That's odd. For, this should be an idealist's distinction, not available to the realist. Yet, the distinction between the infinite and indefinite goes back at least to Descartes and Leibniz – realists both. As I said my original application does nothing to address this anomaly. In fact, it magnifies it.

Part II goes historical in order to solve the problems about assertability and anachronism. I outline the Descartes-Leibniz dialectic about the infinite/indefinite distinction and demonstrate how each position reflects larger philosophical positions. I then demonstrate how Kant's two versions of assertability and his version of the infinite/indefinite dichotomy emerge naturally from his attempts (first pre-critical, then critical) to resolve a tension in Leibniz's philosophy.

In light of this demonstration, Part III adapts the Kripke-model interpretation to represent that dichotomy. I then refine that formal semantics in order to depict Kant's signature critical doctrines. That, in turn, shows why Kant's critical version of the infinite/indefinite distinction is denied to his pre-critical predecessors.

Part IV turns the tables. I use what we have learned about Kant and modern semantics to address a pair of issues in the contemporary critical foundations of logic.

## 1. Kripke models and the antinomies

First a brief precis of basic Kripke semantics, then a rapid capsule of Kant's Antinomy.

### *1.1. Kripke models*

Recall that intuitionistic logic invalidates excluded middle and whatever follows from it. Kripke (1965) provided a formal semantics for first-order intuitionistic logic. Given a first-order language, L, an intuitionistic Kripke model, *M*, for L is an ordered triple <G, K, R> (the frame) together with a domain U, a domain function ψ, and an evaluation function φ. K is the set of nodes, G (∈K) is the base node (the 'actual situation'), and *R* is the (transitive and reflexive) accessibility relation on K. Each node, H, has a domain, Ψ(H) (⊆U), with the condition that H*R*H' implies ψ(H) ⊆ ψ(H'). At each node, H, the interpretation function,φ, gives the extension at H, $\varphi(H,P^n)$, for each n-ary predicate letter, $P^n$ of L. Once again we require that HRH' implies that φ(H, Pn)⊆ φ(H', Pn).

Some propositionally compound formulae hold at a node H (we say 'are forced at H') in virtue of the local situation at that node. Thus

(1) φ(H, A&B) = 1 iff φ(H,A) = φ(H,B) = 1
(2) φ(H, A∨B) = 1 iff φ(H, A) = 1 or φ(H, B) = 1.

But the others require looking at the collection of accessible nodes:

(1) φ(H, A→B) = 1 iff for all H'(∈K) such that H*R*H', φ(H',A)= 0 or φ(H',B)= 1
(2) φ(H, ∼A) = 1 iff for all H'(∈K) such that H*R*H', φ(H', A) = 0.

There's a similar asymmetry between the conditions for '∃' and '∀':

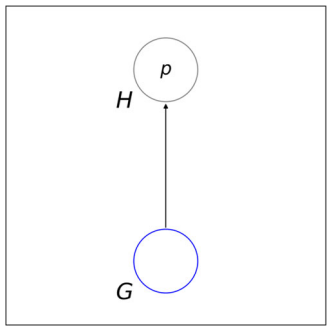


**Figure 1.** The Model $M_1$.

(1) $\varphi$ (H,$\exists$yA($x_1$, … ,$x_n$, y))=1 iff there is a c $\in$ $\psi$(H) such that $\varphi$(H, A($x_1$, … ,$x_n$, c)) =1.

(2) $\varphi$(H,$\forall$yA($x_1$, … ,$x_n$, y)) = 1 iff for each H′($\in$K) such that H*R*H′, $\varphi$(H′, A($x_1$, … ,$x_n$, c)) =1 for every element, c, of $\psi$(H′)[2]

A formula *holds in the model* if it is forced at G. Kripke and others have shown that intuitionistic logic is complete with respect to this semantics.

Here is a simple Kripke model ($M_1$) that invalidates the law of excluded middle, but models its double negation (Figure 1).

You need multiple nodes. Single-noded Kripke models always force excluded middle and, with it, classical logic.

### *1.2. The Antinomy*

As I said, Kant maneuvers his rival realist into a logical contradiction that he (Kant, the critical or transcendental idealist) escapes. Most famous is the first antinomy, where the realist claims that either the material universe extends finitely into space or that it extends infinitely, and then must admit that it can do neither; while Kant's idealist denies the disjunction and says that there is only an ever-expanding series of matter-occupied regions.

However, I'll start with the 'Second Antinomy', concerning the fine-grained analysis of matter. The dogmatist, Kant says, claims that any physical object is either decomposable into indivisible atoms (the Thesis) or else that object has no simple parts and is the sum of an infinite series of ever-finer parts (the Antithesis).[3] Kant then shows that for the realist neither alternative is possible, thus reducing realism *ad*

*absurdum*. The transcendental idealist (Kant himself) escapes this contradiction. For, idealism rejects the disjunction: That object *neither* has atomic parts *nor* is it an infinite telescoping sum. Rather, says Kant, there is simply what he calls a 'regulative task', a 'rule' prescribing a search for ever finer parts and forbidding us to rest with any found division as the final one.[4]

This passage is much-vilified: Kant allegedly has an unclear, overly pictorial notion of infinity; and the whole business of 'regulative' thrust is an unnatural appendage offering a pragmatic, methodological solution to an ontological question. Moreover, the disjunction (finite or infinite divisibility) is a simple logical truth. How can Kant deny it? And, if it is not a logical truth, why is realism – again, a metaphysical, not a logical, view – committed to it? Jonathan Bennett (1974), chapters 7–9, approvingly gathered together these and other accusations. But Kripke models come to the rescue.

Let the variables of L range over components of some piece of matter, say a plank of wood (components that arise from analyzing it into finer and finer parts), and let L have one binary predicate, D(x,y), saying that x and y are such parts and y is strictly finer (comes later in the analysis) than x. Now we formalize the thesis claim about atomic parts by:

$$\exists y \forall x_{\neq y} D(x, y) \tag{1}$$

This says that there is an ultimate legitimate division – and hence an indivisible (simple) part. The Antithesis claim about infinite divisibility becomes:

$$\forall x \exists y D(x, y) \tag{2}$$

It says that the telescoping series of parts of has infinitely many elements.[5]

Allow these formulae as fair renditions of Kant's alternatives, and there is no problem about expressing infinity. It is Formula (2).[6]

And the logical puzzlement dissolves if we associate transcendental realism with classical logic and Kant's transcendental idealism with something like intuitionistic logic: Add the axiom that D is a strict order, and the disjunction (Thesis $\vee$ Antithesis) becomes a classical logical truth, but not an intuitionistic one. Here is a Kripke model ($M_2$) that invalidates that disjunction[7] (Figure 2).

In $M_2$, numbers will stand in for the plank's components. So the domain of the nth node, $H_n$, is {0, 1, 2, … , n}, representing the divisions done up to that node. D (j, k) is forced at a node when j and k are in the domain of that

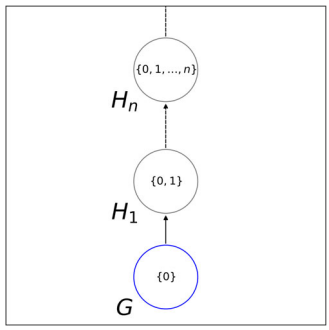


**Figure 2.** The Model $M_2$.

node and j<k. Since no node contains an absolutely smallest division, the clause for negations forces formula (3) at the base node.

$$\sim \exists y \forall x_{\neq y} D(x, y) \tag{3}$$

Similarly, since in no node does each element of the domain have a subsequent element in that domain, formula (4) holds at the base node.

$$\sim \forall x \exists y D(x, y) \tag{4}$$

Notice that this model does validate the formula

$$\forall x \sim\sim \exists y D(x, y) \tag{5}$$

Given any x and any node $H_{N,}$ you cannot negate the claim that there is a division finer than x. Some subsequent node will contain such a division. This says formally – but more importantly our Kripke model depicts graphically – exactly what Kant means by the *regulative* push: We can never rest at a stage knowing that there will be no finer division; but we cannot say we have such a division until we do have it. No artificial appendage here. It stems right from the model.

So intuitionistic logic and its Kripke models nicely interpret Kant's claims about infinite divisibility. But we still need to understand Kant's transcendental idealism in a way that justifies linking it with intuitionistic logic. Transcendental idealism must serve as what Dummett calls a 'philosophical basis' for intuitionistic logic.

As I mentioned, Kant himself goes ontological here: Transcendental idealism, he says, claims that objects are 'nothing outside our representations'. Transcendental realism takes objects to be (mind-independent) things-in-themselves.[8] And the *reductio* of realism is explicitly ontological:

> If the world is a whole existing in itself, it is either finite or infinite. But both alternatives are false … . It is therefore also false that the world (the sum of all appearances) is a whole existing in itself. From this it then follows that appearances in general are nothing outside our representations – which is just what is meant by their transcendental ideality. [A534-5/B506-7]

This sounds like familiar phenomenalism and realism, a fairly standard interpretation of Kant. But a wrong one: It makes Kant sound like a Berkeleyan idealist, something Kant himself denied vehemently.[9] Also, it leaves no room for objects too small or too far or too faint to be sensed. Yet Kant clearly thinks there are such objects.[10] It's logically wrong too: Ontological idealism such as this does not generate intuitionistic logic. It invalidates such intuitionistic tautologies as $\sim\sim(p\vee\sim p)$.[11]

Indeed, Michael Dummett prefers a non-ontological, *assertabilist* semantics for intuitionistic logic. In this semantics, knowledge (i.e. warranted assertability) stands in for the standard reference-based notion of truth.[12] This gives recursive semantic conditions. For instance: To assert $A\vee B$ you must be able to assert A or assert B; to assert $\sim A$ you must show that A is never assertable; to assert $\exists yA(y)$ you must be able to find a c such that A(c) is assertable; and to assert $\forall y$ (Ay) you must have a method converting a proof that c is in the domain to a proof of A(c). Since we often lack the needed proofs and methods, this semantics invalidates bivalence and with it excluded middle.[13]

Certainly, there are good reasons to put Kripke models and assertabilism together. Kripke, in his original presentation, speaks of each node as representing an 'evidential situation', and describes accessibility as modeling the growth of knowledge. Quite specifically, the domain at a node H is to be the set of things that are *known* at H to exist, and Kripke's recursion conditions mimic the standard assertability conditions.

And as for Kant, the *Critique of Pure Reason* is replete with assertabilist thinking (1933). For instance there is the appropriate asymmetry between universality and existence: Universals ('All A's are B's') are true in virtue of a conceptual rule showing that any recognized A is a B; but an existential ('There exists a B') is true only when we can tie such a B to a *current* perception.[14] Recall, our model $M_2$ actually exploits this. And in the Antinomy, specifically, the realist supports the antithesis (the claim of infinite divisibility) by saying 'the existence of an absolutely simple cannot be established by any experience or perception, either outer or inner'. (A437/B465). This reasons from the impossibility of human experience to outright falsity. Realism here is assertabilism with classical logic.

Thus, we have a neat little package of assertabilism as the common denominator linking Kripke models to Kant.

### *1.3. Questions*

But this exercise raises as many questions as it answers. There's the historical anomaly I mentioned: Kant says that the second antinomy's series of divisions continues *ad infinitum* while the first antinomy's series of matter-occupied spatial regions goes on only *ad indefinitum*. Kant's notion of continuability is reserved for the idealist; yet Descartes already deployed the infinite/indefinite distinction as did Leibniz, realists both. (ii) Worse yet, our model $M_2$ serves equally well for the first antinomy, where Kant says that the physical world extends neither finitely nor infinitely into space. Suppose that L's variables now range over parts of physical space (matter-populated regions) and that the predicate D(x,y) means that the region y is farther away from the earth than x. Then formulas (3), (4) still hold, and formula (5) still expresses the idea of a regulative pull to explore farther.[15] So this difference between the series of spatial regions and the series of material divisions seems simply to go lost.

And there are questions about Kant's idealism and the rival realism: (i) if transcendental idealism is assertabilism, why does Kant paint it ontologically? Heyting and Dummett both insist that assertabilism is ontologically neutral.[16] Moreover, why must a realist who holds that empirical objects are things in themselves also hold that the world is a thing in itself? Realism about ordinary empirical things coupled with distrust of abstracta like the sum of all empirical things is a respectable ontological position. (ii) And how can we put assertabilism in the realist's mouth? Or more precisely, why should Kant's critical philosophy favor intuitionistic logic while saddling his non-critical forerunners with classical logic?[17] (iii) Finally, how can we attribute intuitionistic logic and assertabilism to Kant at all? He did not read Dewey, Brouwer, Heyting, Dummett, or Kripke.

But he did read and internalize Descartes and Leibniz. To answer the questions about assertabilism and logic, in Part II I'll trace the dialectic about the infinite/indefinite distinction among Descartes, Leibniz and the two Kants (pre-critical and critical). I will show that each philosopher embedded the distinction in his overall view of knowledge, objects and ultimately truth. Part III will return to Kripke models. I'll use them to explicate Kant's central critical doctrines and thus to answer our questions about his critical philosophy and its version of the infinite/indefinite distinction.

## 2. The infinite and the indefinite: an early modern dialectic

### 2.1. The run-up to Kant

#### 2.1.1. Descartes

Descartes introduced the infinite/indefinite distinction. He did it to contrast our grasp of the notion of God – unquestionably an infinite being – with the way we grasp other seemingly infinite, or at least unbounded, things: the divisions of a plank of wood for instance, or the number of stars in the firmament.[18] Our grasp of those other things – the indefinites – rests on an iterative process of extending our knowledge and on the fact that we can conjure that process's limits only in a negative, allegorical way. By contrast, we grasp God immediately, clearly and distinctly as intrinsically unlimited.[19] This brings in epistemology, ontology and basic semantics.

*Epistemology*: Descartes' notion of constructing geometric curves gives his cleanest epistemology of iterative processes. Here's how it works: Geometric curves

> … can be conceived of as described by a continuous motion or by several successive motions, each motion being completely determined by those which precede; … [I]n this way an exact knowledge of the magnitude of each is always obtainable. (Descartes 1954, *Geometrie*, Book I)

There's a process within a process: The inner process traces a continuous motion. The outer process determines the successive motions in an *effective* way: 'each motion being completely determined by those which precede'. This outer process is designed to preserve the clarity and distinctness of the inner steps. And Descartes' *Geometrie* explicitly tells us which inner motion can be clearly and distinctly grasped: Not the motions generating such curves as the quadratrix and the spiral for instance. These are approximable, but they cannot be traced by graspable continuous motions.[20] On these grounds, Descartes banishes such curves as merely 'mechanical'. The curves that can be clearly and distinctly grasped can be traced in a single continuous motion that presents all the curve's points.[21]

*All* the points! Grasping the whole is a criterion for admissibility, and an effective procedure must preserve that grasp. Star counting and plank dividing are not effective procedures – the next step is not guaranteed or may require choice – and carrying them out fails to provide a grasp of the complete firmament or all the plank components. That's what their 'unboundedness' comes to. The bottom line: True clear and distinct knowledge grasps the whole of its object and comes from an effective process of reasoning applied to whole-grasping, clear and distinct input.

Mathematics is the paradigm, but it is only an instance of Descartes' general 'Method', which applies across the board.

*Ontology*: The *Geometrie*'s lines and curves are forms of spatially extended things; and geometric constructions are abstract forms of mechanical interactions. For Descartes, the essence of an extended thing is to be physically measurable and to interact mechanically with other such things.

*Semantics*: Clear and distinct grasp is veridical, pure and simple. Grasp an object clearly and distinctly, it exists (as grasped). Grasp a situation clearly and distinctly, it obtains. Period.[22]

### *2.1.2. Leibniz*

Leibniz fashioned those same Cartesian epistemological themes – clear and distinct representation, grasp of a whole, even epistemic effective procedures – into a tight-knit pristine metaphysical system. So pristine, however, that he needed special strategies to connect it with ordinary empirical objects and to preserve our empirical knowledge.

*Pristine Epistemology*: A *clear* grasp reveals enough properties 'to recognize the thing represented. My grasp is *distinct* if it serves to tell that thing apart from any real or possible other thing.[23] Good. But the paradigm of clear and distinct cognition is God's grasp of what Leibniz calls a 'complete concept', and that is an infinite thing. The concept of Julius Caesar, for instance, is an infinite array of descriptions that encodes every stage of Caesar's existence.[24] Unpacking that complete concept – the effective combinatory business of conceptual analysis – produces knowledge of particular facts about Caesar; indeed, facts about the whole world. For Caesar's relations with the Rubicon, with his barge, with every plank on that barge, and ultimately with everything and everyone else are encapsulated in that concept.

*Pristine Ontology*: (i) Objects: Only something described by a complete concept is a true *object,* an *individual substance.*[25] As such it will be predicatively complete: it either has or determinately lacks any given property at any given stage. That is its *true unity*. And because it mirrors all relations, it will express the entire world, and the world's causal structure.[26]

(ii) Collections: A description that is incomplete anywhere along the line is *general*: Delete any determination – however trivial – and you describe only the 'aggregate' of things associated with the different ways of filling that slot. Conversely, a composite thing never has a complete concept. That's why true objects are simple, indivisible, *monads*.

A composite thing has only a subjective, 'mental' or 'phenomenal' unity. It is not metaphysically real. We hold its parts and their arrangement

together in our minds; and so its 'very being is also in a way mental, or phenomenal'.[27] To be sure, some aggregates have a seemingly natural unity, those that we unify according to a sortal concept (like person, or ship, or plank), but that is still only a subjective, phenomenal unity, metaphysically second class.

*Pristine Semantics*: This picture gives a rarified assertability semantics. The sentence 'Caesar crossed the Rubicon' is true in virtue of that combinatorial unpacking. Caesar's complete concept *contains* the concept of a Rubicon crossing. This is assertability; for, it is an epistemic state – grasping the unpacked concept – that determines truth.[28] But rarified: Only God could grasp an infinite complete concept; and so it is *God's* knowledge underlying truth here.

This assertability theory does generate classical (propositional) logic! For, as Leibniz says: 'Not true' and 'false' coincide. So 'not false' and 'true' will coincide also. … From these, it is also proved that every proposition is either true or false … . ('General Investigations of the Analysis of Concepts and Truth,' §3, 1966) This is the semantic bivalence that underlies the Law of Excluded Middle.

And Leibniz does connect assertability and ontology: God can assert all those properties that actually inhere in Caesar; so Caesar's individual concept fully expresses his true nature.[29] It is assertability united with reference.

So, this pristine Leibnizian system neatly, coherently combines an epistemic assertabilist framework with the ontology of objects and the world and with classical logic.

*The Tension and Leibniz's Strategies*: The pristine ontology scorns extended objects, bodies: They are divisible and cannot be true substances. The planks of Caesar's barge, for instance, and their arrangement into deck and keel, have only the subjective, 'phenomenal', unity imposed by the viewer. They are not real. Empirical science (studying extended bodies) and indeed mathematics (studying the forms of extension) are thus tainted too.

The pristine epistemology derides perception-based human knowledge. For, perception is always partial, incomplete. Stare as I might at Caesar's barge, my perception reveals neither its full past nor its future. It will not distinguish the perceived barge from others with different pasts or futures. For Leibniz, my perception is neither clear nor distinct, but rather a 'confused' conception of the barge, low on the spectrum of cognitive grasp.

And the semantics disdains empirical truth: After all, we form our empirical concepts from this confused inaccurate experience, and we

form our empirical judgments from these concepts. Such judgments could not possibly accurately depict the way things truly are metaphysically.

In a nutshell: Sensory perception is confused, never veridical, and incapable of apprehending real objects.

But Leibniz was scientist and mathematician as much as metaphysician, with no wish to deride, scorn or disdain physical reality and perception-based empirical knowledge. So his philosophy is replete with strategies to reconcile the human and the pristine pulls. Here are two:

Sometimes he tried to infuse the mechanical and mathematical science of phenomena with metaphysical grounds from the pristine realm, appealing to certain 'forms of indivisible natures' as the causes of what appears rather than to the corporeal or extended mass.[30]

And sometimes, he tried a sort of *reduction*: he tried to show how empirical bodies are ultimately *composed* of monads: A physical object, he says, is repeatedly divisible, until ultimately one reaches its underlying 'monad'. 'The monads', he says, 'are the true atoms of nature; in a word, they are the elements of things'.[31]

This reduction is, I think, the key to Leibniz's own view of the infinite/indefinite distinction. That wooden plank, to be sure, has only a phenomenal unity. But it is a 'well-founded phenomenon';[32] it rests on an underlying monad whose nature dictates and unifies the stages of any division. Thus one can – God can – divide it infinitely down to that monad.[33] But the firmament has no such primary monad. In tracing *its* extent, we are left to our own devices; and Leibniz agrees with Descartes that for us that extent is at best indeterminate.[34]

### *2.2. Kantian strategies*

Note that those Leibnizian strategies are both God-dependent: God, only God, can grasp those 'metaphysical forms of individual natures' in the infusion strategy. God, again only God, can analyze that plank down to its founding monad. Much the same holds for Leibniz's other strategies. Kant admired Leibniz's systematic metaphysics; but for Kant and many of his contemporaries, divine knowledge was an unacceptable metaphysical ploy.[35] So the Leibnizian tension remained. Kant's attempts to reconcile the empirical and metaphysical poles was a central theme throughout his philosophy, and efforts at reconciliation particularly shaped his evolving views about material division and about the extent of the firmament.

Material division was the topic of his 'Physical Monadology' of 1756, where he proposed quasi-punctual 'physical' monads that fill a space in

virtue of their repulsive forces. He took up the issue of the extent of the universe in his final pre-critical work, the *Inaugural Dissertation* of 1770. There he adopted an elaborate avatar of Leibniz's pristine philosophy in a valiant attempt to reconcile the human and pristine poles. I'll look briefly at the *Dissertation*'s reconciliation strategy and its treatment of spatial extent. Then I will tell you about the *Critique of Pure Reason*'s full-blown system and Kant's treatment there of both division and extension.

The *Inaugural Dissertation*: Leibniz's opposing metaphysical *pulls* are now opposing cognitive *poles*. Kant posits two faculties: on the one hand a *Sensory Faculty* (passively received and finitely limited perceptions together with perceptually derived concepts like those of barge, river and motion) and an active *Intellectual Faculty* (with abstract concepts) on the other; both within a single human knower.

He elevates the sensible faculty epistemologically: Sensory perceptions now are 'clear and distinct' intuitions.[36] Semantically too: Empirical predications like 'Caesar crossed the Rubicon' are assertabilistically *true* conceptual containments between sensory-based concepts.[37] But ontologically it still limps: Sensory perceptions do not present authentic objects; and the sensory faculty cannot ground existence. So we have sensible assertability without reference. Indeed, *existence* and other metaphysical notions (*substance, necessity, causality*) belong strictly to the intellectual faculty. Metaphysically adequate knowledge and truth belong to the intellect.[38] So sensory perception, though clear and distinct, is formally but not metaphysically veridical.

As for the firmament: On the one hand, there's no finite end to it; it must be an infinite totality, and thus empirically ungraspable. But Kant offers a Leibniz-style infusion from the intellect to the sensible faculty. With sensibility alone we cannot achieve infinite measurements; we cannot reach the edges of the world. But that's o.k., Kant says: where our sensibility only approximates, our intellect completes. We cannot grasp the world-whole sensibly (intuitively), but no worry, the intellect kicks in and gives us the requisite grasp.[39]

The *Critique of Pure Reason*: Kant soon concluded that this inter-faculty cooperation fails. I'll turn to his reason for this conclusion shortly, but first I'll lay out the *Critique of Pure Reason*'s new conciliatory strategy.

The *Critique* elevates the sensory faculty to a full-service knowledge provider in its own right, the faculty of Understanding. 'Full service' because our human, sense-based knowledge enterprise has its own account of knowledge, its own robust ontology and its own referential semantics.

Epistemologically, sensory perception is now fully objective and veridical. Yes, Caesar only *sees* his barge and its planks cross the river, and yes, the sortal concepts *barge* and *plank* and *river crossing* organize his perception. Indeed, Kant allows no perception without a sortal concept, but now such conceptual organization is objective.[40] Moreover, those heretofore 'metaphysical' concepts (necessity, substance, existence, etc.) are now domesticated in the Understanding. Kant calls them 'schematized categories'. Their job is to show that all such combinations of experience and organization are indeed objectively justified.

And the empirical ontology no longer limps: Caesar's barge, its plank, the river, and even the event 'Caesar's river crossing' – spatiotemporal though they be – these are now full-fledged objects. So sensory perception (empirical intuition) is now indeed ontologically veridical. What we see is what there truly is. Thus the forms of perception, space and time, are the forms of reality. This is Kant's 'empirical realism'.

To be sure, Kant, like Leibniz, distinguishes truly unified objects from mere aggregates. Indeed, he goes fully Leibnizian: An actually existing object must be *predicatively complete*![41] A compound thing about which we know there will be unanswerable questions – something with undeterminable predications – such a thing will be a mere aggregate. Since the sensation is received from an external source, Kant's point is that if you sensibly perceive the plank then you are assured that it is there to answer any question you may ask about it, even if your current perception leaves the questions unanswered.[42] If you merely conceptually contemplate that plank, you have no such assurance.

And the empirical semantics is robust: Caesar's judgment about his barge is *true* in virtue of his sortally organized perception. Existential judgments will be true in virtue of such perceptions; but universal judgments will not rest on actual perception. This is a humanized Leibnizian assertabilism; and it is now fully referential.[43]

Kant calls this humanized Leibnizian system the 'standpoint of the Understanding';[44] it comprises the epistemic, ontological and semantic norms that govern empirical discourse. And it connects Kant's assertabilism, ontology and logic together in a historically responsible way.

Let me point out that this standpoint shapes Kant's critical distinction between the *indefinite* extent of the material universe and the *infinite* divisibility of matter:

First the first antinomy account of the *extent of the universe*: To depict Kant's reasoning suppose we're a scientific task force seeking more and more distant matter-occupied regions of the universe and regularly

reporting our findings. At the point when we report a series of n such regions we have grasped a large but complete object – the amalgam of all those regions. Perhaps we cannot survey it with the naked eye, it is nonetheless surveyable; and that is enough to count as a full-fledged object.[45] However, we cannot claim that there is an n+1st such region, and a yet larger object. We must wait to see whether we receive the needed perceptual evidence. Maybe we'll find such a region in time for the next report, maybe we will not. Indeed, if there is no further such region, we will wait forever and never know that. Similarly, we cannot now claim that we will find a next such region and one after that and one after that, forever. We have no 'effective procedure' taking us from the observation of one region to the observation of the next, from one large object to yet a larger one. We do not even know if there will be the next region. This is true, of course, for any n. Moreover, we cannot jettison our spatial skins to view the whole thing, nor our temporal skins to see how the search will look from the end of time.

Epistemologically, that sensory receptivity and that possibly infinite wait give empirical science unanswered questions (e.g. will there be a next matter-filled region) and sometimes unanswerable general questions (is this the last such region, will there be infinitely many).[46] We thus grasp that world-whole 'only in concept, never as a whole, in intuition'.[47] Ontologically, this means that the world-whole fails the test of complete determination; it is not an object at all. Semantically, as humanized assertabilists, we must deny both that the world-whole has an outermost region and that it extends infinitely.

Next, *material division*. Now we're exploring the components of that wooden plank. Once again Kant says that the collection of revealed subparts is not finite. His reasoning: Right now we grasp the plank, sortally, as a piece of wood. As we probe more we'll reveal its cellular components; and as we probe yet further we will get to the components of those cells, and – Kant will say – even more basic components of those components. (There might well be an amalgam of finer components as we proceed. Cells consist of more basic things.) But, Kant says, we will never come to a dead end with no possible further division. Space's mathematical divisibility guarantees that. Similarly as we go finer and finer we will never find a resolution with nothing but empty space.[48] For, that would preclude matter at grosser resolutions. Moreover, we'll also never hit a level with only uninterrupted matter. That's because, according to Kant, the 'attractive force' holding the parts of matter together works only in empty

space.[49] So there's no finite stopping point. There's always a pattern of matter and interstices. For Kant matter is spongy all the way down.

On the other hand, we know that the telescoping collection of divisions is not infinite. Those patterns are sortally described –organized parts of an object of a given type – so as we go finer, we encounter a sortal and another and another all the way down.[50] Yet at each point in the process the assertion that there exists a further division will not yet be assertable. Perception, recall, requires sortals, but we must wait for (empirical and theoretical) science to provide the appropriate next sortal concept. The series of divisions is not only forever unfinished; it is forever innovative. There is, again, no effective procedure moving us forward. Thus we cannot yet say that the 'next' division is sortally organized and can never say that the series is 'organized to infinity'.[51]

So, once again the world-whole (now the sum of all divisions) will be neither finite nor infinite. And once again, it is only 'an *aggregate*', not an object, 'never whole'.[52] Indeed the series of divisions is not only forever unfinished; it is forever innovative. Once again there's no effective procedure to move us forward.

Now here's the difference between material division and spatial extent: We are uncertain at each report of spatial extent whether we have come to the end of the road with no further regions to be discovered. We proceed *ad indefinitum*. But no such uncertainty plagues the analysis of matter; there's no question of empty space forever or uninterrupted matter. The analysis goes on *ad infinitum*.

## 3. Kripke models and the critical turn

I want now to use Kripke models to precisify this Kantian version of the infinite/indefinite distinction and to distinguish Kant's version from that of his realist predecessors. First I'll model the Kantian distinction and then generalize the modeling to interpret the main lines of Kant's critical philosophy. That, in turn, will show why the distinction is denied to the realist.

### *3.1. Two Kripke models*

Model $M_2$ depicts Kant's view about division: forever searching; yes, always finding; but still refraining to assert the existence of the next component. A node represents an evidential situation verifying those and only those levels of resolution so far documented. Yes, we expect matter for a finer

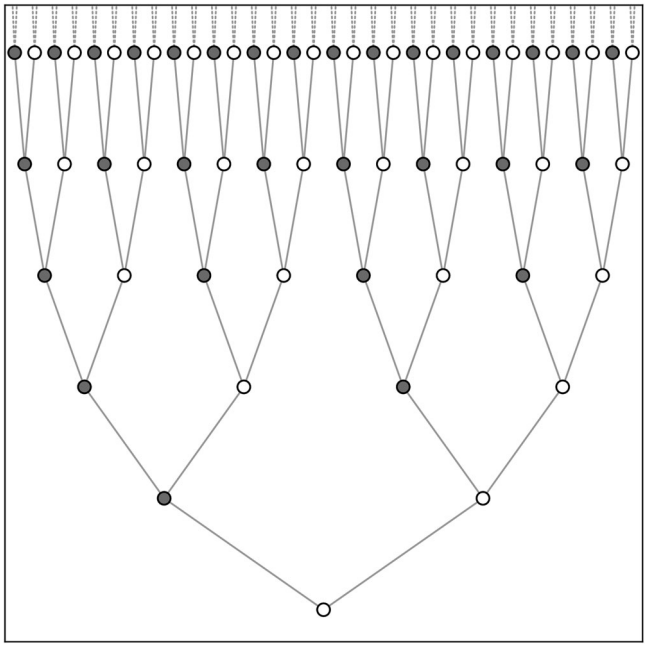

**Figure 3.** The Model $M_3$.

resolution; but *sans* a sortal concept for things that fine, we cannot assert that the level exists. The series of divisions unfinished: No nodal domain contains the full series.

But Kant's picture of matter-occupied space recommends a different model. We always face two possibilities for our next report: yes we'll encounter a further matter-filled region, no we will not. So, we contemplate Figure 3, the frame of a different Kripke model, $M_3$.[53]

Nodes reflect reports. Dark ones indicate finding a further matter-populated region, clear ones say no. D(x,y) says informally that the region represented by y is more distant from the earth than the one represented by x. Formally, again, natural numbers stand in for matter-occupied regions, and n>m indicates that the n-region is further away than the m-region. Each node presents an intuitable object – the region of matter-occupied space amalgamated from those regions so far discerned – and no node contains the full collection of such regions, the world-whole.

$M_3$ forces formulas (3) and (4), Kant's denials that the physical world extends finitely or infinitely into space. It also forces formula (5), signifying regulative pull.

Now the models' different frames precisely render Kant's distinction between the two types of series. $M_3$ depicts a series that continues *ad*

*indefinitem*, where we never know whether we have reached a terminus or if we ever will. $M_2$ shows a series, no less regulative, no less non-infinite and non-finite, but where nevertheless at each point we confidently expect a next. This series continues *ad infinitum*.

Now to Kant's signature doctrines – Reason, synthetic *a priority*, empirical realism, transcendental idealism – and to show why the realist is denied Kant's version of the infinite/indefinite dichotomy.

### 3.2. The standpoint of reason

This is the standpoint of mathematics and Kant's 'transcendental philosophy'.

Mathematicians prove a theorem about e.g. triangles by abstracting from the accidental features of some found, drawn or even imagined triangle and concentrating on features it shares with all other triangles. At this abstract level, Kant sees no possible infinite wait or eternal indeterminacy. He says 'we can demand and expect nothing but assured answers' to every mathematical question.[54] Such epistemic optimism gives mathematics its own ontology and its own semantics and thus dictates a different standpoint from that of the Understanding.

Semantically, optimism supports single-noded Kripke models and classical logic. Ontologically, that, in turn, generates complete mathematical objects. For, recall, a single-noded Kripke model collects together the entire domain, and supports excluded middle (full determination) for every element of that collection. Together this epistemology, ontology and classical semantics constitute the 'standpoint of Reason'.

'Transcendental philosophy' examines empirical science's philosophical and semantic underpinnings and constraints. It justifies the schematized categories, and it shows that the series of extensions and divisions provide tasks and not facts. This metatheoretic enterprise also frees us from receptivity. 'The key', as Kant says 'is within us'. So, here too we have optimism and classical logic, the standpoint of Reason.

Ontologically, we actually get infinite things. Transcendentally we have an abstract representation of space that is 'given' as infinite.[55] We have an effective procedure for extending this abstract space without end: take a line and extend it in thought. So, when our variables range over abstract regions of space, we can assert formula (2).

In treating models $M_2$ and $M_3$, a modern transcendental task, we also adopt Reason's standpoint: No node contains the full set of divisions or

extensions, but the models themselves do have that forbidden set: the global domain **U** that helps us define individual nodal domains. $M_2$ and $M_3$ are infinite objects. Moreover, Kripke's and most other completeness proofs use classical reasoning.

One could tinker with the meta-theory here. There were intuitionistic versions of Kripke models and completeness theorems.[56] But we want none of that. We want the meta-theory just exactly as Kripke sketched it. The content of the nodes remains intuitionistic; the model itself is viewed classically, from Reason's standpoint.

The models $M_2$ and $M_3$ force all the same sentences of L. They are empirically indistinguishable; only transcendentally do we see their different shapes. So, for Kant, the infinite/indefinite distinction properly belongs to transcendental philosophy; and we formulate it from Reason's standpoint.

### *3.3. Synthetic* a priori

Mathematics studies the spatiotemporal forms of perceivable objects. Its judgments are synthetic, because they're tied to perception; they're known *a priori*, because those forms are ubiquitous. Transcendental principles, like causality, are similarly synthetic and *a priori*. With a tweak, the Kripke semantics expresses this synthetic *a priority*.

(A) *Analytic/Synthetic* Our empirical language, L, has no individual symbols. It cannot express such thoughts as: 'there will be third level components, and when we find them they will be finer-grained than the second level ones'. But we can expand L to a language L′ that does express those thoughts.

Syntactically, we add countably many individual symbols, say, '0′′,' '$\sigma$(0)', … , '$\sigma^n$(0)', etc. ['$\sigma^n$(0)' abbreviates 'σ(… (σ(α)) …)' with n iterations of σ].

Informally: 0 represents our sortally distinguished starting object – e.g. that wooden plank – and $\sigma^n(0)$ stands for the sortally distinguished component uncovered at the $n^{th}$ level analysis. Formula (6) represents the claim 'The $j^{th}$ component is more refined than the $k^{th}$.'

$$D(\sigma^k(0),\ \sigma^j(0)) \tag{6}$$

Formula (7) represents the specific claim 'The n+1st component is more refined than the nth'.

$$D(\sigma^n(0),\ \sigma^{n+1}(0)) \tag{7}$$

For the material division model we'll have '0' denote 0, and '$\sigma^{n+1}(0)$' denote n+1 in general. Then to see if an atomic sentence like (6) is forced at a node H, we would look at the referents of $\sigma^k(0)$ and $\sigma^j(0)$ to see if the pair $<\sigma^k(0), \sigma^j(0)>$ is in the extension of D at H. That will happen if j<k.

Clearly (7) is not forced at any node H before the n+1$^{st}$. For, at least one of $\sigma^{n+1}(0)$ and $\sigma^n(0)$ will have no referent at such an H, and $< \sigma^{n+1}(0), \sigma^n(0)>$ could not be in D's extension there. This seems not quite right. Certainly, we can say such things from the start about the nth and n+1st components. Indeed, in Kantian terms (7) should count as an analytic truth.

This is the business of 'free logic': giving formal semantics for true sentences containing non-referring expressions. So we want a free intuitionistic semantics here, one that expands the conditions for elementary predication at a node as follows:

A predication, $P(t_1, \ldots, t_k)$, holds at a node H if

EITHER

(i) all of the terms $(t_1, \ldots, t_k)$ have denotations in the domain of H and $<t_1, \ldots, t_k>$ belongs to the extension of P at H

OR

(iia) There is at least one node H* such that all of $(t_1, \ldots, t_k)$ do have denotations at H*

AND

(iib) For any node H' in which all of $(t_1, \ldots, t_k)$ do have denotations, the n-tuple $<t_1, \ldots, t_k>$ belongs to the extension of P at H'.[57]

In a free intuitionistic model $M_2$* with $M_2$'s frame, case (i) says that $P(t_1, \ldots, t_k)$ holds *synthetically* at H; while in case (ii) it holds *analytically* throughout the model. (ii) guarantees (7) in the model $M_2$*.[58]

When we move to the question of the world's spatial extent, and read L as the language of matter-filled regions, then '0' will represent our current farthest matter-occupied region, and $\sigma^n(0)$ will stand for the n$^{th}$ subsequent region to be discovered. (7) now says that 'the n+1$^{st}$ matter-occupied region is more distant from us than is the n$^{th}$'; and once again it will be analytically true.

(B) *A priority*: Our observations are fallible: The extent-of-matter task force might take a mirage for a new region or make a miscalculation. The division-of-matter team could err in identifying that n+1st component. Think of phlogiston and ether. Our modified Kripke semantics straightforwardly accommodates this fallibility: To show that the base node G has a fallible observation, just add a G′ – inaccessible to G – reporting some incompatible observation (a different nth component, for instance). Now the sidelong, incompatible observation is itself receptive;

it too generates a constellation of projections, further observations consistent with G′. We add that constellation to the model. We might err in different ways – posit different components – so we can add more than one non-accessible constellation.

However, some things will persist – must persist – through all of these constellations. Every such constellation will have $M_2$'s frame. In the course of material division each constellation *will have* a node with an n+1st component, an n+2nd component and so on. If we were observing an event, then each constellation would have us eventually discover the cause of that event. Similarly each constellation – each node of each constellation, in fact – will force mathematical truths. No matter what the n+1$^{st}$ component might be, if our language accommodated geometry, then Euclidean geometry would characterize the space in its interstices.[59]

For Kant, *a priori* propositions sometimes earn that status because they are conditions for the possibility of experience, sometimes they simply reflect the nature of human perception; but in either case, we depict this via their *a priority* by invariance in a collection of constellations.

### *3.4. Transcendental idealism and empirical realism*

I say 'depict', not 'explain'. Nothing here explicates Kant's transcendental proofs of the *a priority* of mathematical or categorial principles. Nothing denotes his notions of transcendental object or self that figure into those proofs. On the other hand, we *can* model the restriction of those arguments to the forms of *human* intuition.

Kant prominently contemplates the possibility of finite (receptive) beings, whose forms of intuition differ from our own.[60] We depict the abstract possibility of such beings by adding new galaxies of constellations where even the principles of mathematics do not hold – one or another non-Euclidean geometry might hold – and where the categories will be differently schematized. Our expanded Kripke semantics adds additional clusters of constellations, additional galaxies, without a blink.[61] This, in fact, represents the *synthetic a priority* of mathematical claims and categorical principles: Though true throughout the constellations representing human intuition, they will not hold throughout the model, so they are not analytic.

But it is important to keep in mind that no matter how many inaccessible epicycles we might add (in which synthetic a priori principles fail), a Kripke model makes true those and only those formulas forced at its one base node. Truth is truth at the base node. This depicts Kant's empirical realism. The base node resides in the human, Euclidean galaxy. Truth

'for us' is truth, period. It is true simpliciter that our plank is wooden, and if it is square its angles sum to 360°.

This also precisely depicts Kant's transcendental idealism. Base node truths are 'ideal' because that human galaxy is one-among-many. Conformity to the human forms of intuition is, in some large metaphysical sense, contingent. That's idealism. It is 'transcendental', because in noting that contingency we are doing transcendental philosophy. Models of all shapes belong to transcendental not empirical discourse. Their frames and expansions belong to the transcendental ontology, along with the abstract notions of object, truth and self that Kant uses in his transcendental arguments. And, to repeat, when we model the transcendental discourse in which we discuss these things, we will use single-noded models. For, that discourse is governed by the standpoint of Reason.

### *3.5. The critical turn and transcendental realism*

Mathematics impacts empirical knowledge and empirical objects. We calculate daily with arithmetic and measure with geometry. In an appropriate language, mathematical truths show up in every human node of each one of our Kripke models. Transcendental philosophy has a similar impact. In a language of events and causality, each 'human' constellation forces the eventual discovery of a cause for any perceived event. Moreover, reflection on our models (a task of transcendental philosophy) provides the regulative push for continued research into the ever larger and the ever smaller parts of the world.

Thus, one is tempted to accept the Understanding's ontology of empirically knowable objects while maintaining the epistemic optimism and classical logic that goes with mathematical and transcendental thought. Succumb to that temptation and you will admit the entire abstract domain as itself a legitimate object. Your single-noded Kripke model will collect together the entire domain into a single object and support excluded middle (full determination) for it. So you will have a modern humanized assertabilism together with classical logic and its world-whole. This is the dogmatist of the Antinomy. But this is the primrose path to antinomy: By logic this global object, the world-whole, is finite or infinite. That is, the disjunction of Formula (1) and formula (2) holds. But the Understanding's arguments still prevail; and so their negations (3) and (4) hold as well. Contradiction!

The 'critical' aspect of Kant's philosophy warns us to avoid this trap: to separate the two standpoints, and to exercise what Kant calls the 'discipline' needed to avoid the temptation of mixing them.

Now who would succumb to this temptation? Well, Kant did in the *Inaugural Dissertation*, where the intellect completes our limited empirical knowledge. The intellect's semantics is single noded, its logic classical. And so in the *Dissertation* the empirical world is not open-ended, there are no unreachable bounds or unplumbable depths. And this, Kant now thinks, is a mistake: The intellect's additions are not legitimate empirical objects; if you think they are, you fall into the Antinomy. So Kant here refutes his own prior strategy of infusion.

Leibniz, the father of reduction and infusion, is clearly in the same camp. As is Descartes. Think of his optimistic application of pure thought to all scientific questions. Reasoning alone, he says, applied to clear and distinct (ultimately intellectually derived) inputs will answer any question.[62]

All of these pre-critical positions assume that there is only one theoretical enterprise (detailing the world). Metaphysical reasoning and its objects will be part and parcel of the tool kit for that enterprise. Methodologically, Kant calls that dogmatism. Epistemologically, it is optimism. Semantically, its models are all single-noded; its logic always classical. Ontologically, it is realism: There is one domain in which the pristine and the empirical, the mathematical and the metaphysical all reside together. There is no room here for inaccessible nodes, constellations or galaxies. And that's what lets Kant reduce this amalgam *ad absurdum*.

A brief word about using these formal methods to interpret Kant. Some commentators think that Kant's pre-Fregean syntax rendered him unable to express infinity and generated his philosophical attitudes about infinite domains.[63] In fact, however, Kant is closer to modern logical syntax than is often recognized.[64] But the issue here is not *expressing* infinity, nor is it about actual versus potential infinity. All Kant's infinities are potential – even the given infinity of metaphysical space. The intuitionistic reading of the ∀∃ quantifier combination in formula (2) enforces this. The question is rather how to distinguish between a fully graspable versus a forever innovative potential infinity. Kripke's intuitionistic models – whose domains grow but, in our case, remain finite– precisely capture Kant's notion of the forever innovative.[65] That is why Kant's distinction between the 'given' infinity of metaphysical space (formula 2) and the regulative infinity of divisions and expansions (formula 5) belongs only to this transcendental discourse of multi-noded models. It is denied to the single-node realist.

This also answers our question about Kant's version of the infinite/indefinite distinction. The distinction depends on different shaped Kripke models, and consequently it is denied the realist. That's what I set out to show.

## 4. The critical view of logic

Now let us turn the tables and use my modeling of Kant's critical philosophy briefly to address two modern philosophical questions in the foundations of logic.

### *4.1. Logic, ontology, and cognition*

Kant's critical philosophy held that one human cognitive activity (the enterprise of empirical science) is governed by a finitary standpoint. I want to say that the Kantian notion of a standpoint addresses the issue of whether and how logic is grounded in ontology or in cognition. Contemporary philosophers have pondered this question at least since Husserl's *Logical Investigations*.

My use of formal models showed that the human standpoint generated a free-intuitionistic logic. But as we saw, logic went hand in hand with ontology, epistemology and cognition, and for that matter the pragmatics of assertion. This is what a *standpoint* is: a systematic philosophical unity in which all of these components are equal partners.

This is perhaps the deepest Leibnizian legacy. His pristine metaphysics seamlessly melded all those ingredients. I led with epistemology – clear and distinct was the head notion – but I could have started from any one of the ingredients and get the others. Kant strove to save that philosophical unity. He adjusted the ingredients of Understanding's standpoint to do so: the world cannot be an object; our grasp of it cannot be intuitive; truth must be humanly assertable. The non-classical empirical logic came with all of those together – with the cognition, the deep ontology, the pragmatics. That is its Kantian grounding.

I said that the empirical enterprise is *governed* by its standpoints. The standpoint is a set of *norms*: Norms governing what we may assert and when; what counts as knowledge and how to get it; what to expect (in fact predict) of objects; (an object must be ready to yield up all it nature and properties); how we are to reason (what inferences to accept, which to abjure). To work from the finite human standpoint is to be governed by all of these norms, that special logic among them. This, I think, is as much a source of that logic's normativity as anything else. Modern intuitionistic logic, is embedded in a full-scale finitary standpoint, a view of objects, assertion and truth.

This holds no less for Brouwer's mathematical intuitionism than for Kant's transcendental idealism. Brouwer, I believe, does mathematics

from the finite human standpoint, thus joining logic with ontology and cognition.

### *4.2. The logic of the metalanguage*

'Critical' also means that other cognitive projects – mathematics, transcendental philosophy, for instance – are governed by a different standpoint, Reason's standpoint; one that differs from the Understanding on each axis of norms.

This is not relativism. Mathematics and transcendental philosophy, for Kant, guide empirical science, but are ancillary to it; unequal partners in displaying reality. We need, in fact, the critical discipline to know when Reason's enterprises work together with science and when they must be kept apart.

In §3.2, I argued that for Kant this means that the logic of one's metatheory need not – should not – be the same as that of the object theory it serves. The Understanding and its logic govern empirical discourse, but Reason and its logic govern the metatheory. This issue of a separate meta-logic is hotly debated these days.

I daresay Brouwer did not accept Kant's view. He favored 'Understanding' all the way up. I will not here adjudicate between Kant and Brouwer. I will only say that Kant's philosophy gives a clear, coherent and 'critical' precedent for current fans of a separate meta-logic.

## Notes

1. This part corresponds roughly to Posy (1992). Posy (1983) presented a preliminary version.
2. The quantifier conditions are generally done using assignments and satisfaction conditions. See Kripke (1965) or Dummett (2000) for full formal definitions of a Kripke model.
3. See A434/B462, A435/B463.
4. See A509/B537.
5. The Antithesis statement doesn't mention infinite divisibility, but in his 'Observation on the Antithesis '(A439/B467) Kant equates the Antithesis with the claim of infinite divisibility. I am using 'divisibility' and 'analysis' interchangeably. For, as Kant says, the division is 'in thought' (A434/B462).
6. To be sure Kant did not have modern formal languages, and I'll say something briefly later about the legitimacy of this formal translation.
7. I first applied Kripke semantics to the second antinomy in a talk, I gave at the conference, 'Saul Kripke, Philosophy, Language and Logic, A Celebration of his work', CUNY Graduate Center, January 25–26, 2006.
8. A491/B519.

9. See *Prolegomena*, AK IV, 375 (2002a).
10. See A226/B273.
11. Ontological phenomenalism naturally grounds non-bivalence. It leaves undetermined propositions as truth-valueless. (Otherwise it could not distinguish between an object t whose construction determines ∼Qt, and one whose construction leaves Q undetermined.) But phenomenalism gives no ground for anything but minimal deviation from the classical truth tables. Thus if p is truth-valueless, so too will be ∼p, ∼∼p, and (p∨∼p). Thus ∼∼ (p∨∼p) will be truth-valueless.
12. Dewey's (1938), I believe, is the origin of the term 'warranted assertability'. Dummett systematically takes warranted assertability as the ground for intuitionistic logic. See for instance, Dummett (1978).
13. The recursion conditions were first set out in Heyting (1930) in terms of mathematical proof. A more general version speaks simply of assertability in an epistemic situation. See Posy (1992).
14. See A225/B272.
15. Indeed, this is precisely the model I used in my early interpretations of the first antinomy. See for instance Posy (1983, 1991a, 1992).
16. See Heyting (1966) chapter 1 and Dummett (2000) chapter 7.
17. Putnam (1978, 26–27) suggests that a certain mapping between intuitionistic and classical logic unites assertabilism with classical logic. But this doesn't work for general assertabilism or for Kant. See Posy (2015).
18. Here's the *locus classicus*:

> We must not try to dispute about the infinite, but just to consider that all in which we find no limit is indefinite, such as the extension of the world, the divisibility of its parts, the number of the stars, etc. … Thus because we cannot imagine an extension so great that we cannot at the same time conceive that there may be one yet greater, we shall say that the magnitude of possible things is indefinite. And because we cannot divide a body into parts which are so small that each part cannot be divided into others yet smaller, we shall consider that the quantity may be divided into parts whose number is indefinite. And because we cannot imagine so many stars that it is impossible for God to create more, we shall suppose the number to be indefinite, and so in other cases. … And we shall name these things indefinite rather than infinite to reserve to God alone the name infinite, first of all because in Him alone we observe no limitation whatever, and because we are quite certain he can have none, and in the second place in regard to other things, because we do not in the same way positively understand them to be in every part unlimited, but merely negatively admit that their limits, if they exist, cannot be discovered by us. (*Principles of Philosophy*, Part I, Principles 26, 27 (AT VIIIA.15; Descartes [1970], I, 229–230))

Paul Franks pointed out to me that Ernst Cassirer attributed a similar distinction to Nicholas of Cusa (see Cassirer 1963, 14) . However the dialectic that I am tracing here clearly began with Descartes.

19. I am indebted to Anat Schechtman's reading of Descartes on these issues. See, in particular, her (2018, 27–44). However, I do not follow her on all details.
20. '[T]he *quadratrix, and similar curves*, which really do belong only to mechanics, and are not among the curves that I think should be included here, since they must be conceived of as described by two separate movements whose relation does not admit of exact determination. ' [Descartes, *Geometrie*, Book II.]

    To be sure, Descartes' properly 'geometric' curves have algebraic descriptions; but that is a consequence of the possibilities of construction, not a free-standing criterion of admissibility. See Henk Bos (2001).
21. Their algebraic expressions ultimately describe processes for producing those points.
22. See, e.g. *Meditations on First Philosophy*, Meditation II. Descartes had no inductively defined logic. The Port Royalists provided a 'Cartesian' logic.
23. Leibniz, 'Meditations on Knowledge, Truth and Ideas' (1969a).
24. See Leibniz, 'Discourse on Metaphysics', §13 (1969b).
25. Leibniz, 'Discourse on Metaphysics', §8 (1969b).
26. Leibniz, 'Discourse on Metaphysics', §9 (1969b).
27. Leibniz, *New Essays II,* ch. 2, section 7.
28. Indeed, in 'General Investigations of the Analysis of Concepts and Truth' [1686] (his attempt to give a 'formal semantics') Leibniz comes close to standard assertabilist truth conditions: A true judgement is one 'which can be proved' (§130). A false one is such that it can 'be proved that, however long an analysis is continued, it cannot be proved that it is true. ' (§57). *Universality* is just a question of unpacking a relation between a pair of general concepts, to see if one is included in the other. Existence, by contrast, is hard to prove: '[E]xistential propositions … cannot be proved except by using infinities or by a continuous analysis involving an infinite number of facts, namely, only from a complete notion of an individual, which involves infinite existence.' (§75.) See also §74.
29. This is what Leibniz means when he says: 'God … in seeing the individual notion or *haecceity* of Alexander, sees in it at the same time the basis and the reason for all the predicates which can truly be affirmed of him … ' ('Discourse on Metaphysics' §9, 1969b) Ironically, Leibniz is reverting to those very same scholastic notions that Descartes and his fellows reviled.
30. 'Discourse on Metaphysics', §18 (1969b). This is developed in his theory of 'living force' (*vis viva*). See the *Specimen Dynamicum* (1969d).
31. *Monadology*, §3 (1969c).
32. See, e.g. his letter to Remond of February 11, 1715 (G. III, 634–635, L. 658–660).
33. See Leibniz (2001, 233–235).
34. See Leibniz (1949) II, XIV.
35. … I will explain natural appearances as though they came from the constitution of nature; I cannot call upon God. For that would mean putting aside all philosophizing. (*Metaphysik Mrongovius AK* 1997, 774)
36. See *Inaugural Dissertation*, AKII, 395.
37. *Inaugural Dissertation*, AKII, 397.
38. *Inaugural Dissertation*, AKII, 395.

39. … even if these co-ordinations [of a world-whole (CP)] could not be sensitively conceived, they would not, for that reason, cease to belong to the understanding. It is sufficient for this concept that co-ordinates should be given in some way or other, and that they should all be thought as constituting a unity. (*Inaugural Dissertation,* II, 391–392)

    See Posy (Forthcoming) for a fuller treatment of the *Inaugural Dissertation*.
40. See A103.
41. See A571-2/B599-600 and A573/B601.
42. An empirical object has no in-principle unknowable residue, no aspect which is intrinsically beyond our human ken.
43. See Posy (2000).
44. See Bxviii–xix, n.
45. Descartes and Leibniz speak also of counting stars, Kant speaks only of matter-occupied regions. For, he is concerned with making the regions into objects on their own.
46. See A480/B508. Kant is speaking here of global statements. Singular predicative questions about existing empirical objects can in principle be answered.
47. A518/B546.
48. See A514/B542.
49. See, *Metaphysical Foundations of Natural Science*, Chapter II, Proposition 7. AK 4, 512 (2002b).
50. At some point the pattern might remain the same (in a fractal way, perhaps). In this case there would be a most fundamental sortal. Kant will not allow that. Such a sortal would give a pattern characterizing matter itself. It would give additional structure and content to the notion of matter beyond its *a priori* properties. A similar observation could be made about repeating patterns. I do not know whether Kant would have considered that possibility.
51. A526/B554.
52. A525/B552.
53. The model $M_3$ has a more delicate formal definition than $M_2$. Here it is:

    $\mathbf{K} = \{ < a(0) \ldots a(k) > \mid k \in \boldsymbol{N},\ a(i) \in \{0, 1\},\ 0 \leq i \leq k\}$

    For all $a$, $a(0) = 0$

    $\mathbf{G} = \{a(0)\} = \{0\}$

    $< a(0) \ldots a(k) > \mathbf{R} \quad < a(0) \ldots a(k), a(k+1) >$

    $$\Psi( < a(1) \ldots a(k) > ) = \{0, \ \ldots, \ m\}, \quad m = \sum_{i=0}^{i=k} a(i)$$

    $\mathbf{U} = \boldsymbol{N}$

    For all $< a(0) \ldots a(k) > \in K$ , $\varphi( < a(0) \ldots a(k) > , F) = \{ < n, m > | n, m \leq$

    $$\sum_{i=0}^{i=k} a(i), \ n > m\}$$

    That is For all H $\in K$, $\varphi$(H, D(n, m)) = T iff $n > m$

54. A480/B508.
55. B39–40.
56. See Veldman (1976); de Swart (1977); and de Swart and Posy (1981).
57. §5.3 of Posy (2007) explores a variety of semantics for free intuitionistic logic. Posy ([1982] 1991) provides completeness proofs.
58. A Kantian nicety: The expressions $\sigma^n$ (0) are *conceptual* expressions until the point (if there is one) that they come to denote. Analyticity always rests on conceptual grounds. Posy (1991b) relates this to Russell's notion of logically proper names.
59. This, by the way, depicts the difference between what Kant calls 'regulative' versus 'constitutive' *a priori* principles: The regulative ones are realized in constellations, the constitutive ones at every node.
60. See (A27/B43), B72, B138–139, B145, B149, A230–1/B283, and A286/B342.
61. The added constellations and galaxies of nodes are not formally idle appendages. The broad scope of condition (ii) for analyticity assures that they will affect the evaluation of wff's at the base node, G.
62. Kant addresses Descartes' realism in the fourth Paralogism of the *Critique of Pure Reason*. This passage explicitly works out the result of Descartes' mixing the empirical and transcendental enterprises.
63. See for instance Russell's *Principles of Mathematics*, sections 435 ff, and Friedman (1992), chapter 2.
64. In particular, Kant's philosophy underwrites the primacy of singular predication and the reality of relations, two ways in which traditional (and Leibnizian) logical syntax are pre-Fregean.
65. E.W. Beth's earlier semantics for intuitionistic logic requires a constant domain throughout a model (see Beth 1953, 1959, §145). This renders that semantics unfit for modeling Kant. Lopez-Escobar (1980, 1981) formally compare Beth and Kripke models. I will discuss the heuristic and philosophical differences between these semantic systems on another occasion.

## Acknowledgements

I thank the editors for their patience and encouragement and for their helpful comments on an earlier draft. I also want to thank Saul Kripke for the work that inspired this paper and for his friendship over the years.

## Disclosure statement

No potential conflict of interest was reported by the author.

## Funding

Thanks to the Israel Science Foundation for its support. Early research for this essay was supported by ISF grant 914/02. Grant 1491/14 supported later research and writing.

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